

A Cobb–Douglas Approach to Resource–Constrained Time–Cost Tradeoff Problems in Project Crashing

Industrial Mathematics Week

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13 July, 2026

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Introduction

Project crashing is a schedule compression technique to meet late deadlines, creating an economic **time–cost tradeoff** [Caglayan and Yigit 2024](#).

Resource-Constrained Project Scheduling Problems (RCPSPs) are the de-facto model for real-world scheduling [Artigues et al. 2026](#):

Project graph $(I, \mathcal{E})$ of activities $i \in I$ with precedence $\mathcal{E}$

Resources set $\mathcal{K}$ with capacity R_k ; requirement $r_{i,k}$

Schedule start times s_i and end times e_i satisfying feasibility

Objective minimize $\mathcal{G}(s_{n+1})$

Key Challenge

RCPSPs are **NP-hard** [Blazewicz et al. 1983](#) due to resource constraints, making optimization method selection essential.

Brucker's α | β | γ notation [Brucker et al. 1999](#) classifies RCPSPs by:

α Resource environment (renewable, non-renewable)

β Activity characteristics (prec, temp, ...)

γ Objective function (makespan, time-cost tradeoff, ...)

Three approaches to decision variables:

1. **Time-based:** direct schedule control (duration, binary indicators) [Kelley Jr. 1961](#)
2. **Mode-based:** binary selection of execution “modes” (MRCPSP) [Brucker et al. 1999](#)
3. **Resource-based:** direct resource allotment (RIP/RACP) [Hartmann and Briskorn 2022](#)

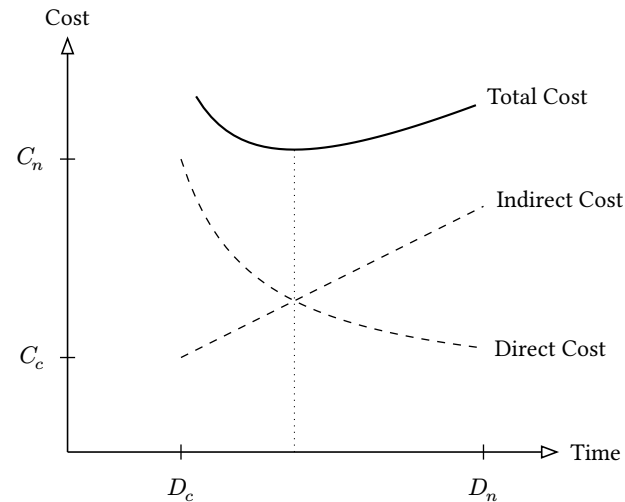
The choice of decision variables determines the objective γ , and is one of the most important aspects of RCPSPs.

Time--Cost Tradeoff Problems

Resource-constrained Time--Cost Tradeoff Problems (RC-TCTPs) reduce project costs while minimizing makespan [Hosseinpour et al. 2023](#).

Direct costs labor, materials

Indirect costs leaseholds, penalties, overheads



The interplay of cost types yields a nontrivial tradeoff. TCTPs often employ **multi-objective** functions or scalarizations [Agarwal et al. 2013](#).

Exact Methods:

1. **Mixed-Integer Linear Programming** (MILP) — best for mode-based decisions
2. **Constraint Programming** (CP) — more flexible, interval variables [Laborie 2009](#)
3. **Branch and Bound** (B&B) — widely applicable to MILP and CP
4. **CP-SAT / SAT solvers** — state-of-the-art results [Artigues et al. 2026](#)

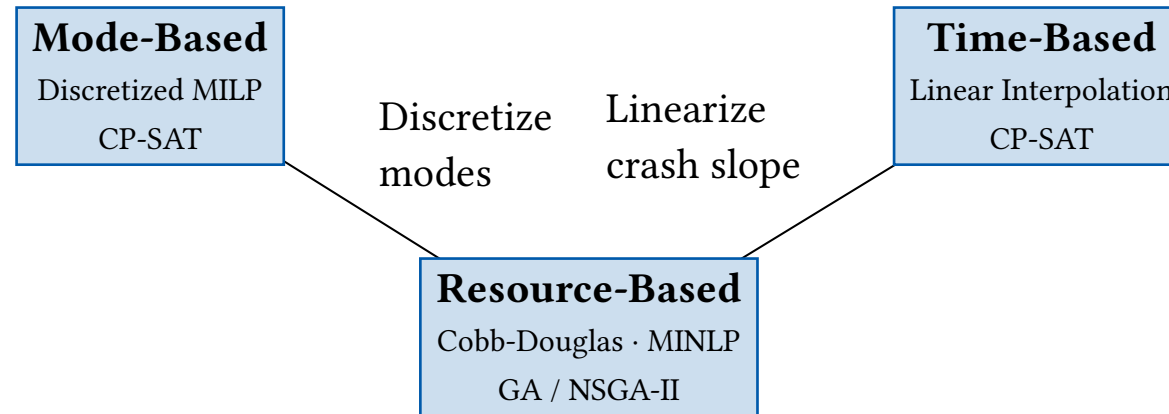
Non-Exact Methods:

1. **Genetic Algorithms** (GA) — most notable metaheuristic [Hartmann 1998](#)
2. **Particle Swarm** (PSO) and **Ant Colony** (ACO) [Merkle et al. 2002](#); [Zhang et al. 2005](#)
3. **Machine Learning** — promising but computationally expensive

Our Approach

We employ **GA** (NSGA-II) for the non-linear model and **CP-SAT** for the linearized/discretized models.

Models



Three formulations of increasing simplification, all exploiting the **Cobb-Douglas production function**:

$$Y(L, K) = AL^{\alpha}K^{\beta}$$

where α and β are labor and capital elasticity [Cobb and Douglas 1928](#). We build on Shen et al. [Shen et al. 2016](#) who applied Cobb-Douglas to project crashing with crew labor and equipment.

Given a baseline schedule with activities I and resources K :

Baseline effective duration (8 hours/day):

$$d_{i,k}^{(0)} := \frac{W_{i,k}^{(0)}}{8U_{i,k}}$$

Baseline schedule:

$$d_i^{(0)} := \max_{k \in K} d_{i,k}^{(0)}, \quad e_i^{(0)} = s_i + d_i^{(0)}$$

Baseline cost:

$$z_{i,k}^{(0)} := d_{i,k}^{(0)} \cdot U_{i,k} \cdot 8 \cdot r_k = W_{i,k}^{(0)} \cdot r_k$$

$W_{i,k}^{(0)}$ baseline work effort (unit-hours)

$U_{i,k}$ baseline daily allocation (units)

r_k hourly wage (\$/hour)

Two crashing strategies: **overmanning** ($x_{i,k}$) and **overtime** ($\tau_{i,k}$).

Naive duration:

$$d_{i,k} = \frac{W_{i,k}}{(8 + \tau_{i,k})(x_{i,k}U_{i,k})}$$

Implies $z_{i,k} = z_{i,k}^{(0)}$ (no extra cost) – unrealistic!

Cobb-Douglas duration with $A_{i,k} = U_{i,k}^{1-\alpha}8^{1-\beta}$:

$$d_{i,k} = \underbrace{\frac{W_{i,k}}{8U_{i,k}}}_{d_{i,k}^{(0)}} \cdot \underbrace{\left(\frac{1}{x_{i,k}}\right)^\alpha}_{\text{overcrowd}} \cdot \underbrace{\left(\frac{8}{8 + \tau_{i,k}}\right)^\beta}_{\text{overwork}}$$

Expanded cost:

$$z_{i,k} = \underbrace{W_{i,k}r_k}_{z_{i,k}^{(0)}} \cdot \underbrace{x_{i,k}^{1-\alpha}}_{\text{overcrowd}} \cdot \underbrace{\left(\frac{8 + \tau_{i,k}}{8}\right)^{1-\beta}}_{\text{overwork}} \cdot \underbrace{\frac{8 + \frac{r'_k}{r_k}\tau_{i,k}}{8 + \tau_{i,k}}}_{\text{extra wage}}$$

The Cobb-Douglas approach captures **inefficiencies** in overmanning and overtime crashing:

“**overcrowd**” diminishing returns from extra workers

“**overwork**” diminishing returns from extra hours

“**extra wage**” overtime premium $r'_k \geq r_k$

Temporal:

$$s_j \geq e_i + \delta_{ij}, \quad \forall (i, j) \in E_{\text{FS}}$$

(and SS, FF)

Duration bound:

$$d_{i,k} \geq \gamma \cdot d_{i,k}^{(0)}$$

Resource:

$$\sum_{i \in I} x_{i,k} U_{i,k} \cdot \mathbb{1}\{s_i \leq s_j < s_i + d_{i,k}\} \leq U_k^{\max} \quad \min \sum_{i \in I} \sum_{k \in K} z_{i,k} + c_{\text{late}} \max(0, s_{n+1} - T_{\max}) - c_{\text{early}} \max(0, T_{\max} - s_{n+1})$$

Crashing: 3 task types — unstated (I_1), completed (I_0), partial ($I_0^C \cap I_1^C$)

Multi-objective:

$$\min \left(s_{n+1}, \sum_{i \in I} \sum_{k \in K} z_{i,k} (x_{i,k}, \tau_{i,k}) \right)$$

Single-objective (bonus-penalty):

Decision variables: $s_i, x_{i,k}, \tau_{i,k}$. This is a **MINLP**.

The non-linearity motivates a **metaheuristic** approach — Genetic Algorithm (GA) [Hartmann 1998](#).

Chromosome structure ($2|I| \cdot |K| + |I|$ genes):

Genes 1 to $|I| \cdot |K|$ overmanning multipliers $x_{i,k}$

Genes $|I| \cdot |K| + 1$ to $2|I| \cdot |K|$ overtime addends $\tau_{i,k}$

Remaining genes activity order priority $\in [0, 1]$

Evolution:

1. Uniform initialization between bounds
2. Tournament selection
3. Simulated binary crossover (SBX)
4. Polynomial mutation
5. Implemented in pymoo (Python)

Serial Scheduling Scheme: Build schedule from priority order, push start times forward, shift on resource conflicts. Cost acts as fitness score. Generalizes to multi-objective via **NSGA-II**.

Partition overmanning $[1, x_{\max}]$ into M intervals and overtime $[0, \tau_{\max}]$ into N intervals:

$$x_{i,k}^{(m)} = 1 + \frac{m}{M}(x_{\max} - 1), \quad \tau_{i,k}^{(n)} = \frac{n}{N}\tau_{\max}$$

Each pair $(m, n) \in \mathcal{M}$ is a **mode** with **precomputed** duration and cost:

$$d_{i,k}^{(m,n)} = d_{i,k}^{(0)} \cdot \left(\frac{1}{x_{i,k}^{(m)}} \right)^{\alpha} \cdot \left(\frac{8}{8 + \tau_{i,k}^{(n)}} \right)^{\beta}$$

Decision variable: $\xi_{i,k}^{(m,n)} \in \{0, 1\}$ (mode selector)

Uniqueness: $\sum_{(m,n) \in \mathcal{M}} \xi_{i,k}^{(m,n)} = 1$

Objective (linearized):

$$\min \left(s_{n+1}, \sum_{i,k} \sum_{(m,n)} z_{i,k}^{(m,n)} \xi_{i,k}^{(m,n)} \right)$$

This is a **Mixed-Integer Linear Program** (MILP).

$$z_{i,k}^{(m,n)} = z_{i,k}^{(0)} \cdot \left(x_{i,k}^{(m)} \right)^{1-\alpha} \cdot \left(\frac{8 + \tau_{i,k}^{(n)}}{8} \right)^{1-\beta} \cdot \frac{8 + \frac{r'_k}{r_k} \tau_{i,k}^{(n)}}{8 + \tau_{i,k}^{(n)}}$$

Reformulate using **interval variables** [Laborie 2009](#) with start, end, size, presence attributes, and cumulative / exactly **global constraints** [Beldiceanu et al. 2014](#).

$$\begin{aligned}
 \min \quad & \left(s_{n+1}, \sum_{i \in I} \sum_{k \in K} \sum_{(m,n)} z_{i,k}^{(m,n)} \cdot \text{pres}(\xi_{i,k}^{(m,n)}) \right) \\
 \text{s.t.} \quad & \text{exactly} \left(1, \left\{ \text{pres}(\xi_{i,k}^{(m,n)}) \right\}, 1 \right), \forall i, k, \\
 & \text{size}(\xi_{i,k}^{(m,n)}) = d_{i,k}^{(m,n)}, \\
 & \text{pres}(\xi) = 1 \leftrightarrow \text{start}(\xi) = s_i, \\
 & e_i = \max_{k \in K} \sum_{(m,n)} \text{pres}(\xi) \cdot \text{end}(\xi), \\
 & s_j \geq e_i + \delta_{ij}, \forall (i, j) \in E_{\text{FS}}, \\
 & \text{cumulative} \left(\text{task} \left(\left\{ \xi, x_{i,k}^{(m)} U_{i,k} \right\} \right), U_k^{\max} \right), \forall k, \\
 & \text{pres}(\xi_{i,k}^{(0,0)}) = 1, s_i = s_i^{(0)}, \forall i \in I_0, \\
 & s_i \geq T_0, \forall i \in I_1, \\
 & s_i = s_i^{(0)}, \forall i \in I_0^C \cap I_1^C.
 \end{aligned}$$

Solved via **CP-SAT** (Google) – hybrid of CP propagation and SAT conflict-driven clause learning (LCG) [Krupke 2026](#). Integer domains via scaling/rounding:
 $d^{m,n} = \lceil 10^{\ell_1} \cdot \dots \rceil, z^{m,n} = \lfloor 10^{\ell_2} \cdot \dots \rfloor$.

Assume **direct costs are linearly related** to duration reduction:

Linear cost:

$$z_{i(d_i)} = C_i \left(d_i^{(0)} - d_i \right) + z_i^{(0)}$$

Crash slope from Cobb-Douglas extremal values:

$$C_i = \begin{cases} \frac{z_i^{\text{crash}} - z_i^{(0)}}{d_i^{(0)} - d_i^{\text{crash}}} & \text{if } d_i^{(0)} > d_i^{\text{crash}} \\ 0 & \text{if } d_i^{(0)} = d_i^{\text{crash}} \end{cases}$$

Resource slope (same idea):

$$V_{i,k} = \frac{u_{i,k}^{\text{crash}} - u_{i,k}^{(0)}}{d_i^{(0)} - d_i^{\text{crash}}}$$

CP-SAT program:

$$\begin{aligned} \min \quad & \left(s_{n+1}, \sum_{i \in I} C_i \left(d_i^{(0)} - \text{size}(d_i) \right) + z_i^{(0)} \right) \\ \text{s.t.} \quad & d_i^{\text{crash}} \leq \text{size}(d_i) \leq d_i^{(0)}, \\ & s_i = \text{start}(d_i), \quad e_i = \text{end}(d_i) \\ & s_j \geq e_i + \delta_{ij}, \forall (i, j) \in E_{\text{FS}} \\ & \text{cumulative} \left(\text{task} \left(\left\{ d_i, u_{i,k}^{(0)} + V_{i,k} \left(d_i^{(0)} - \text{size}(d_i) \right) \right\} \right), U_k^{\max} \right), \forall k \\ & s_i = s_i^{(0)}, \text{size}(d_i) = d_i^{(0)}, \forall i \in I_0 \end{aligned}$$

Experiments

Real commercial project construction dataset from **Integrated Decision Systems Consultancy** (IDSC), Singapore [Integrated Decision Systems Consultancy Pte Ltd 2023](#).

Activities $|I|$ activities (procurement, substructure, superstructure, envelope, interior, commissioning, closeout)

Resources $|K|$ resources with wages r_k and capacity $U_k^{\max}$

Baseline 344 days projected completion

Overtime $r'_k = 1.5r_k$ (standard) [Hamandia-Güldenbergl 2004](#)

Standardized parameters:

$\alpha = \beta = 0.7$, $T_0 = 20$, $T_{\max} = 250$, $c_{\text{early}} = \$2000/\text{day}$, $c_{\text{late}} = \$5000/\text{day}$

Methods:

1. Resource-Based: NSGA-II (population 1000, max gen 500)
2. Mode-Based: CP-SAT with ε -constraints ($\Delta\varepsilon = 4$ days, $\Delta = 0.1$)
3. Time-Based: CP-SAT with ε -constraints

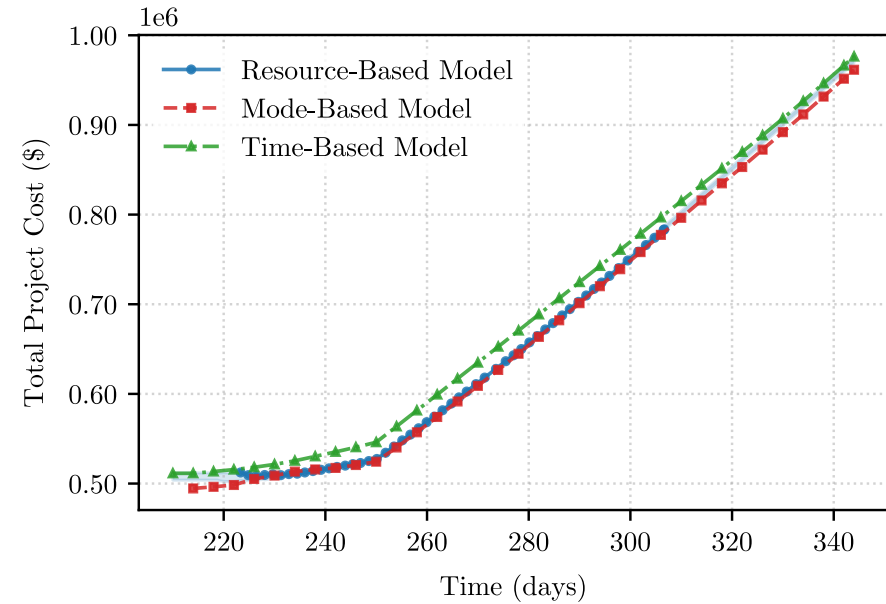
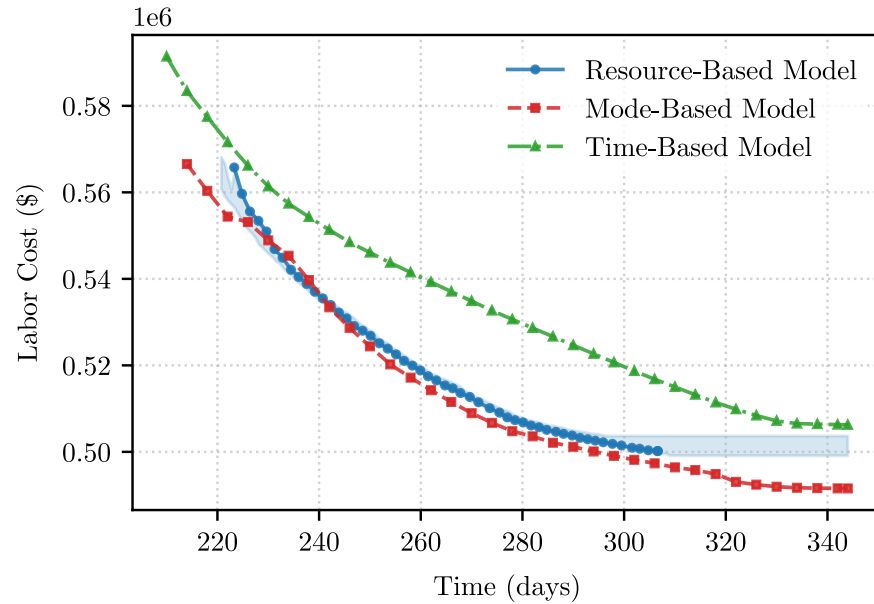
Multi-Objective Results

Metric	Resource-Based	Mode-Based	Time-Based
Contribution (% of Pareto)	8.6%	13.5%	0.4%
Hypervolume	71.9% ± 0.4%	76.6%	58.9%
Min Makespan	221.6 ± 1.6 d (\$564.5k)	214.0 d (\$566.5k)	210.0 d (\$591.4k)
Min Cost	\$501.6k ± 1.6k (302.1 d)	\$491.5k (344.0 d)	\$506.2k (344.0 d)
Solve Time	840.1 ± 64.2 s	1006.3 s	11.3 s

Key Findings

Mode-Based dominates optimality (76.6% hypervolume). **Time-Based** is unbelievably fast (11.3s for entire Pareto front — 0.32s/schedule). Resource-Based is an uninteresting middle ground.

Pareto Fronts & Total Cost



Left: Mode-Based captures the best schedules at any deadline. Resource-Based struggles with the full Pareto front (NSGA-II cannot evolve minimal crashing). **Right:** Adding bonus/penalty ($T_{\max} = 250$) covers short-completion cost rises and punishes late schedules.

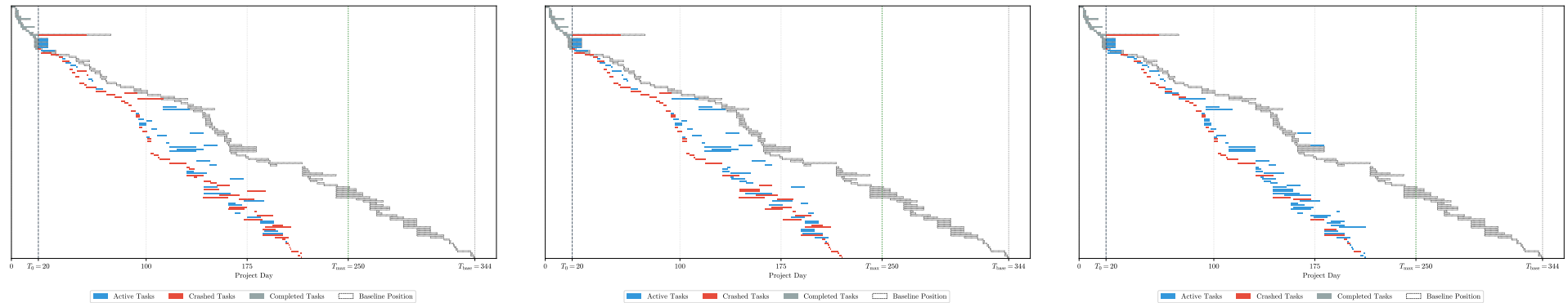
Single-Objective Results

Metric	Resource-Based	Mode-Based	Time-Based
Makespan (s_{n+1})	217.14 ± 0.65 d	220.33 d	213.00 d
Target Margin ($T_{\max} - s_{n+1}$)	32.86 ± 0.65 d	29.7 d	37.0 d
Labor Cost	$\$561,967.72 \pm \$1,072$	$\\$553,426.94$	$\$584,970.19$
Bonus	$\$65,711.31 \pm \$1,298$	$\$59,340.00$	$\\$74,000.00$
Total Cost	$\$496,256.41 \pm \$1,094$	$\\$494,086.94$	$\$510,970.19$
Solve Time	1275.8 ± 6.9 s	301.2 s	2.16 s

Key Findings

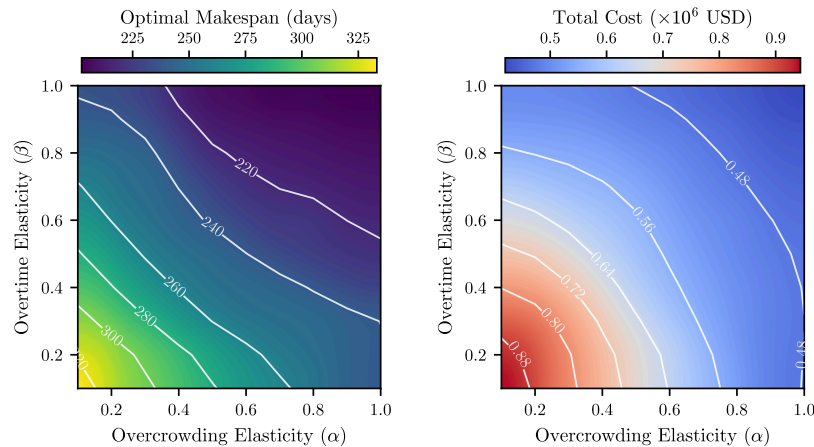
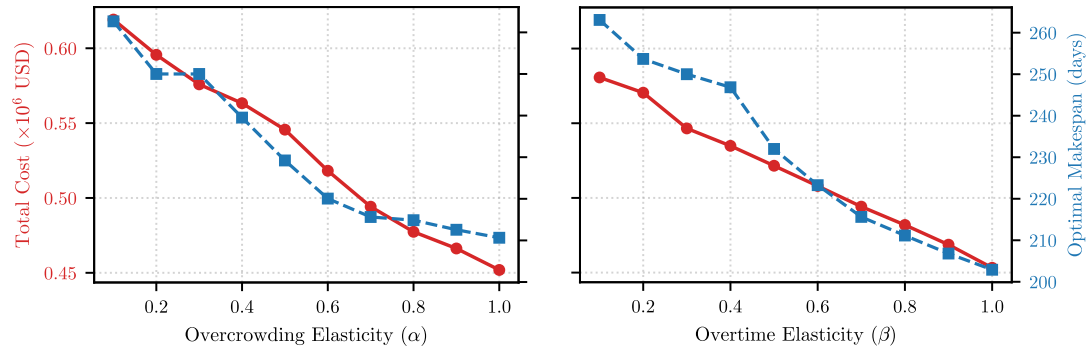
Mode-Based achieves the best total cost. **Time-Based** provides the most extreme crashing (213 d) and is 140× faster than Mode-Based. GA is now the slowest (no NSGA-II parallelization benefit).

Gantt Charts



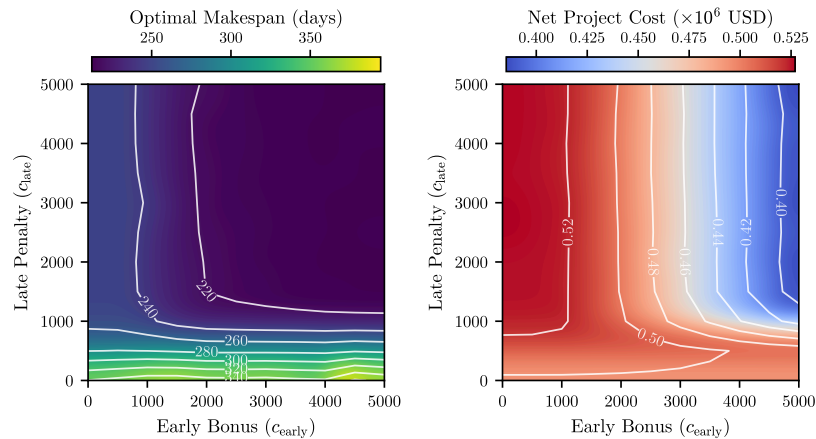
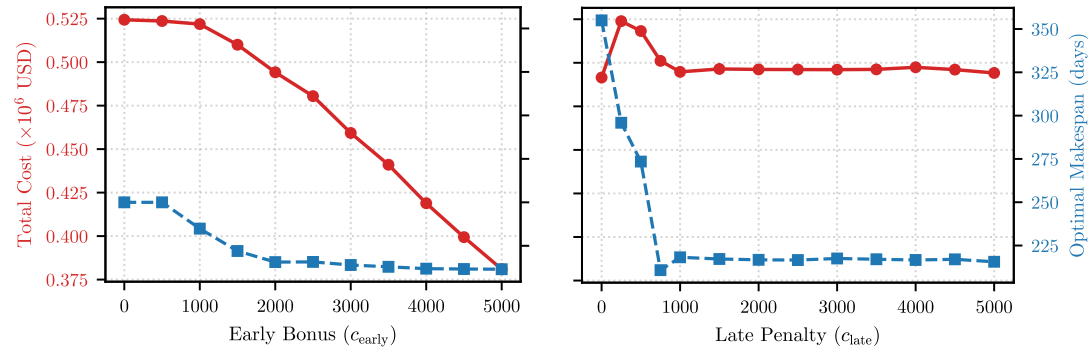
Red = crashed activities. All three models exhibit **emergent critical-path behavior** — selectively crashing bottleneck activities rather than uniformly. Some reordering shows the power of well-defined precedence relations.

Sensitivity: Cobb--Douglas Elasticities



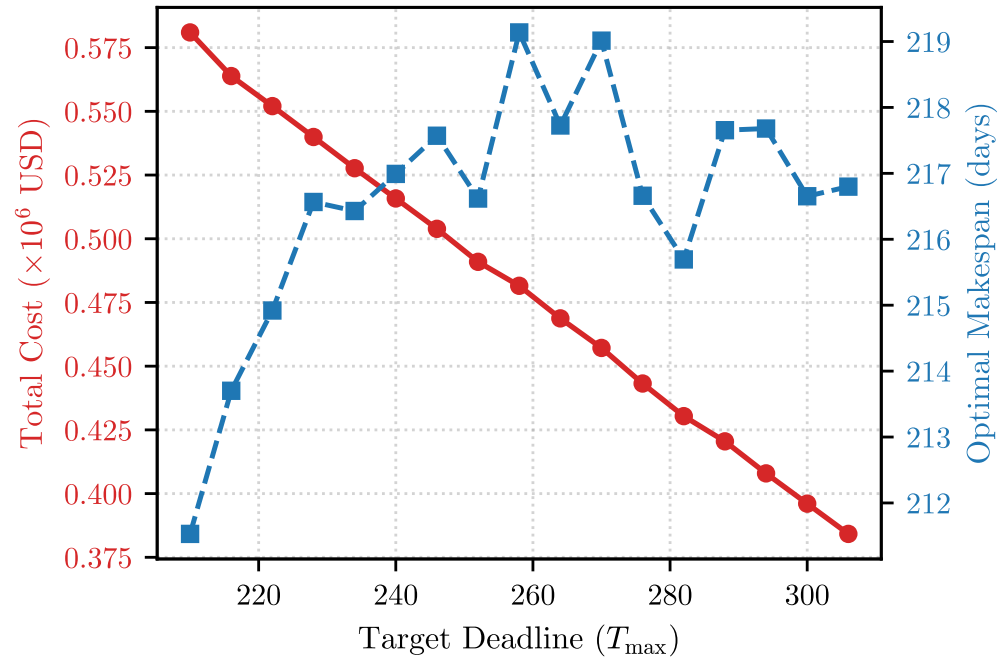
1. **Negative relationship:** Both α (overcrowding) and β (overtime) inversely affect total cost and makespan.
2. **Lateness threshold:** Schedules are **late** (over $T_{\max} = 250$ d) when $\alpha < 0.3$ or $\beta < 0.3$ – capturing inability to accelerate under extreme inefficiency.
3. **One-factor linearity:** Despite Cobb-Douglas being non-linear per activity, aggregation across the schedule is **approximately linear** in α and β – justifying the time-based model's linearity assumption.
4. **Two-factor nonlinearity:** The heatmap reveals the true nonlinearity – contour lines curve and densify at lower values.
5. **Hypothesis:** Time-based model may perform poorly when α and β vary across activities.

Sensitivity: Bonus--Penalty Parameters



1. **Early bonus** (c_{early}): Greater bonuses **incentivize extreme crashing** — drastically decreasing both total cost and completion time.
2. **Penalty** (c_{late}): More complex. Acts as a **soft constraint** for on-time completion.
3. **Critical region**: $c_{\text{late}} \in (0, 1000)$ — tradeoffs between rising labor costs and tardiness penalties are most convoluted (densely packed contours).
4. **Big- M regime**: Beyond \$1000, penalties force on-time schedules that settle at lower total cost via earlier unpunished makespans.
5. Bonuses remain **gradual but clear**; penalties are **non-trivial** in the critical region.

Sensitivity: Project Deadline $T_{\max}$



Surprising Findings

1. **Linear cost:** Total project cost is **almost perfectly linearly related** to the target deadline — corroborating the time-based model's linearity assumption.
2. **Non-linear crashing:** Yet this cost is achieved through a **non-linear crashing strategy** — the Cobb-Douglas model meets every target deadline, but the optimal makespan trajectory is non-trivial.
3. **Robustness:** The model consistently meets every deadline tested, demonstrating the **reliability** of the Cobb-Douglas approach across project horizons.

Conclusion

Contributions: Three novel RC-TCTP formulations using the **Cobb-Douglas production function**, relaxing the assumption of known crash costs.

1. **Resource-Based** (original): non-linear MINLP, solved via NSGA-II GA. Faithful to reality but computationally heavy.
2. **Mode-Based**: discretized MILP, solved via CP-SAT. **Best optimality** — 76.6% hypervolume, lowest total cost.
3. **Time-Based**: linear interpolation, solved via CP-SAT. **Best speed** — 11.3s (multi) / 2.16s (single), with acceptable accuracy.

Key Insights:

1. Sensitivity analysis confirms robustness; α/β aggregation is approximately linear, justifying the time-based simplification.
2. All models exhibit **emergent critical-path behavior** — selectively crashing bottlenecks.
3. CP-SAT dominates GA for the discretized/linearized formulations.

Future Work: Dynamic α/β across activities; incorporating bonus/penalty into the optimization; larger-scale benchmarks.

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Thank You!
Any questions?